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Dr. Jongseok Lee
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Dear Dr. Lee,

I am applying for a PhD position at the RUNES Lab. The lab's focus on uncertainty-aware manipulation and field-deployable robotic systems aligns closely with what I have been working on for the past three years, and I would be glad to join its founding cohort.

My master's thesis at Bilkent University, supervised by Dr. Ozgur Oguz, addressed physical-parameter uncertainty in contact-rich manipulation. The resulting paper, **CoRAL** (accepted at RSS 2026), proposes a neuro-symbolic controller with two nested loops. An inner MPPI loop performs reactive control at 1 kHz with force feedback, while a slower outer loop runs online system identification through an LLM: when robot interaction reveals model mismatch, such as unexpected slippage, the LLM updates physical parameters (mass, friction) and the cost structure mid-execution. The system treats uncertainty as a signal that drives adaptation rather than something to be averaged out, and it produced over 50% higher success rates than VLA baselines on novel contact-critical tasks, validated on a Franka Research 3.

The rest of my work approaches the same theme from adjacent angles. **H-MaP** (RA-L 2025, ICRA 2026) combines symbolic planning with continuous motion optimization to handle horizon-level uncertainty in multi-stage tasks. **FViT-Grasp** focuses on 6-DoF grasp detection where calibrated confidence matters as much as raw accuracy.

Beyond papers, I have spent a lot of time getting things to work on hardware. The CoRAL pipeline involves FoundationPose perception, MPPI rollouts, and impedance control across three frequency tiers, and I built and debugged most of it on our lab's Franka. In parallel, I work full-time as an ML engineer at Trendyol, where the distance between a clean offline result and a system that survives production traffic is something I deal with every week. That experience has shaped a strong preference for methods that account for uncertainty by design rather than as a post-hoc correction.

Of the four directions you outlined, manipulation under uncertainty and the engineering of robotic systems for field deployment are where my background fits most naturally. CoRAL, H-MaP, and FViT-Grasp are all manipulation papers, and most of the work behind them sat on the unglamorous side of getting methods to actually run on hardware, tuning controllers, debugging pose estimation, fighting with calibration. What I want from a PhD is to keep doing exactly that, but with uncertainty pushed into the center of the design rather than handled at the edges: building manipulation systems that stay reliable when the world stops cooperating with the assumptions baked into them.

Joining a lab in its founding year is also appealing on its own terms. There is something clarifying about helping define a research program from the ground up, and RUNES Lab is where I would like to do that work.

Thank you for considering my application; I would welcome the chance to discuss it further.

Sincerely,
Berk Çiçek